

HEXAVALENT CHROMIUM Cr(IV) CONTAMINATION EHS PLAN

List of Contents:

1.	PURPOSE AND SCOPE	2
2.	RESPONSIBILITIES.....	3
3.	CR(VI) OCCUPATIONAL EXPOSURE LIMITS	3
4.	WHERE CONTAMINATION COULD BE FOUND	3
5.	ACTIVE VERSUS PASSIVE WORK.....	4
6.	SAMPLING RESULT OF CONTAMINATION	5
7.	PRE-JOB PLANNING	5
8.	GENERAL CONTROL MEASURES	5
9.	EXCLUSION ZONE	6
10.	PERSONAL PROTECTIVE EQUIPMENT	8
11.	CLEANING	8
12.	DECONTAMINATION BEFORE HOT WORK	9
13.	CLEANING/DECONTAMINATION OF TURBINE PARTS DURING OUTAGES PRIOR TO SHIPPING OFFSITE TO GE SERVICE CENTRES OR 3RD PARTIES	9
14.	APPLICATION OF ANTI SEIZE.....	9
15.	PERSONAL HYGENE.....	11
16.	HOUSEKEEPING	11
17.	PROHIBITED ACTIVITIES	11
18.	ENVIRONMENTAL ASPECTS - WATER	11
19.	ENVIRONMENTAL ASPECTS - AIR	11
20.	ENVIRONMENTAL ASPECTS - WASTE	12
21.	ATTACHMENTS.....	12

1. PURPOSE AND SCOPE

- 1.1 The purpose of this Health & Safety Plan is to specify the protection to be provided to employees to prevent exposure to the hazards associated with hexavalent chromium Cr(VI) exposure.
- 1.2 This program is limited to the localized Hexavant Chromium Cr(VI) contamination. It is intended to support work only with Cr(VI) contamination in this form.
- 1.3 Observations on several outages globally with the Gas, Stream and Aero Service lines have showed Cr(VI) contamination where a yellow residue (color can range from off-white/grey to a strong yellow) has been observed on hardware.
- 1.4 A GE Vernova engineering team has determined that the root cause was from the interaction of Calcium containing compounds and heat, reacting with Chromium containing metal alloys. This was predominantly related to where anti-seize was applied, however the mechanism can potentially occur with any compound or chemical that has significant Calcium (Ca) content e.g. Glass wool Insulation, grouting etc

Regardless of the source, it is to be assumed that various hardware on the current GE Vernova (including legacy Alstom) fleet may be is potentially contaminated. Gas, Steam Turbine internals have been impacted.

If outages find suspected contamination, this document will assist in assessing the risk and actions needed.
- 1.5 Because we have not determined the source of the potential Cr(VI), we need to continue to address each situation independently.
- 1.6 This document specifies procedures, engineering controls, work practices and personal protective equipment necessary to control Hexavalent Chromium contamination during operations that produce dust.
- 1.7 The plan applies to all work activities that result in occupational exposure to hexavalent chromium or hexavalent chromium containing materials, including cleaning, inspecting, repairs and transportation.



Figure 1. Examples of yellow residue Cr(VI) accumulation locations on turbine & rotor bolts



Figure 2. Examples of off-white residue Cr(VI) accumulation



Figure 3. Examples of residue Cr(VI) accumulation under insulation

2. RESPONSIBILITIES

2.1 EHS Department:

- Provides guidance on control of Cr(VI) through implementation of this procedure.
 - Be aware of the requirements & regulations in your region with regards to Cr(VI).
 - If IH monitoring is required, after consultation with GE/ FieldCore ensure that exposure assessments and monitoring are performed by or supervised and documented by an approved Industrial Hygienist.

2.2 Site Lead - Supervisors:

- Ensures that this procedure is implemented.
- Ensures the proper management of control measures for engineering, administrative, and PPE are in place.

2.3 Site EHS Representatives:

- Manages the Cr(VI) program at the site and ensures compliance.
- Ensures that all affected employees are notified and receive required training.
- Maintains all documentation and records associated with this program.

2.4 Field based including Craft Personnel

- Be aware of the potential Cr(VI) hazard, while working on turbines.
- Ensures the proper use of the supplied personal protective equipment.
- Comply with given instructions to reduce or eliminate their exposures to hexavalent chromium.
- Participation, when approached, in any exposure sampling for Cr(VI) is deemed necessary.

3. Cr(VI) OCCUPATIONAL EXPOSURE LIMITS

3.1 GE Vernova and Fieldcore apply OSHA Permissible Exposure Limits (PEL) and Action Level (AL) unless there are stricter applicable occupational standards regionally.

3.2 Occupational Exposure Limits

US OSHA (Occupational Safety and Health Administration) has established OELs, which are called Permissible Exposure Limits (PEL).

Personal Exposure Limit for Cr(VI) is 5.0µg/m³
(for an 8-hour Time Weighted-Average (TWA) exposure.)

Action Limit: for Cr(VI) at 2.5µg/m³.

4. WHERE CONTAMINATION COULD BE FOUND

4.1 Not exhaustive, potential for other combustion components and other turbine areas.

4.2 List of the gas and steam turbine internals, the contamination is typically associated with the hot internal (with interaction with Calcium (Ca) containing compounds e.g., anti-seize, cement/grout, AES insulation (Alkaline Earth Silicate), Rock Wool insulation.

- Segment Retaining Pin's other pins and caps
- Shroud block retaining Pin's
- Combustion Liners
- Transition Piece
- Stage One Nozzles
- Turbine Rotor
- Nozzle Support Ring Bolt Holes and associated bolts
- Nozzle and Shroud half-joint hard seals
- Turbine Shell Rail Fits
- Bolting and holes to access hatches to Steam valves. (is the only current known steam hardware impact.
- Thermocouple grouting

Gas Turbines: certain combustion hula seals, nozzles, caps, pins, and turbine rotor bolts.

Steam Turbines: various high temperature locations such as keys, bolting and seals. A small number of cases have been encountered on GE Vernova steam turbines to date on high temperature bolting and other such locations where anti-seize is routinely applied (e.g. keys and guides, packers/shims).

Aero Derivatives: Various high temperature locations such as, bolting and seals, where anti-seize has been applied.

All:

Residues have also been observed on the outer surfaces and other areas of the turbines including internal bucket/blades, piping, which may have a formation mechanism driven by factors other than anti-seize, so the same testing and EHS precautions are recommended if any suspects off-white/grey to bright yellow residues are detected.

5. ACTIVE VERSUS PASSIVE WORK

Active Work is defined as:

*Any task or activity that intentionally or unintentionally **physically disturbs hexavalent chromium-containing material or surfaces**, with the potential to generate airborne dust, fume, mist, or re-suspend settled contamination.*

Key characteristics of Active Work:

Alters the **physical condition, integrity, or location** of Cr(VI)-containing material

Has the **potential to increase airborne concentrations**

Typically requires **engineering controls, PPE, and exposure monitoring**

Examples of Active Work (Cr(VI)):

Grinding, sanding, cutting, drilling, or machining Cr(VI)-containing coatings or alloys
Welding, torch cutting, or thermal processes involving chromium-containing materials
Abrasive blasting or surface preparation
Mechanical cleaning (wire brushing, power tools)
DONOT USE - Dry sweeping or compressed air cleaning of contaminated areas
Removal or disturbance of contaminated insulation, debris, or residues

Passive Work is defined as:

*Any task or activity performed in an area with known or potential hexavalent chromium contamination that **does not physically disturb contaminated materials or surfaces** and is not expected to generate airborne Cr(VI).*

Key characteristics of Passive Work:

No direct contact with or disturbance of Cr(VI)-containing material

Minimal likelihood of **re-suspension or aerosolization**

Typically relies on **administrative controls and basic hygiene practices**

Examples of Passive Work (Cr(VI)):

Visual inspections or walkdowns
Monitoring, instrumentation checks, or data collection
Office or clerical work within a controlled area
Supervision or oversight conducted without contacting surfaces
Transit through contaminated areas following established controls

	TITLE: Hexavalent Chromium Cr(VI) Contamination EHS Plan	DOCUMENT NUMBER: 0807-G1			
		REGION: Global	REVISION: 4.0	DATE: MAR-2026	PAGE: 5 of 12

6. SAMPLING RESULT OF CONTAMINATION

- 6.1 Field based sampling, as detail in Section 19.6 of this document outlines, the method to determine if any suspected contamination contains Cr(VI) If give a qualitative result, i.e., if Cr(VI) is present or not, (rather than a quantitative result, %Cr(VI)) Any positive field test, ie color change of the test reagent indicated Cr(VI) is present and the control detailed in this document should be followed. If there is not color change, a negative result, then the material tested, is unlikely to contain Cr(VI), if in any doubt in tests, repeat the sampling, for confirmation purposes.
- 6.2 Typically, at this time, further laboratory sampling of contamination is not required routinely. There are enough, results to justify the assumption that if there is off-white - yellow residue coated on parts, it contains Cr(VI), and should be treated as such. Field based sampling, detailed in 5.1, is sufficient
- 6.3 Various wipe samples of surrounding surfaces adjacent to the yellow contamination have come back as not containing Cr(VI), confirming the localized nature of the contamination. i.e. the contamination does not spread to surrounding adjacent areas or parts
- 6.4 For potential contamination that is present in an unusual or atypical form* ie not baked on, with loose dust or present as a dust under insulation. Additional sampling may be required after liaison and discussion with the FieldCore Industrial Hygienist.
- 6.5 For testing suspected contamination present under insulation, it is advisable to not use WD40 or similar as this can present a fire risk, and water should be used instead to attain a sample.

7. PRE-JOB PLANNING

- 7.1 Due to the Cr (VI) present on the internal (and possibly on the external section for instance under insulation) of turbines. Prior to all gas, steam and aero turbine work, assume the units are contaminated with the residue. Have the controls ready to be used, (eg WD40, PPE, RPE) if Cr (VI) contamination is found. It should be noted that not all turbines have been affected, and even if a single unit test negative for Cr (VI) it should not be assumed that the other units are not contaminated, and normal visual inspection and testing should occur.
- 7.2 Be sure that all workers with potential exposure to Cr(VI) have been trained on:
 - [Hexavalent Chromium Awareness CRSE- GE-EHS-425\]](#) which is included as part of your Field Services EHS Curricula
 - Awareness of the Hazards associated when working with Cr (VI) and the controls required, to protect themselves.
 - A toolbox talk is usually sufficient for the contamination risk being seen. See Appendix
 - [Business Specific: if determined by your business EHS department, Additional Online Training is available on the GE Vernova Learning & FieldCore LMS that covers some basic awareness of the EHS requirements for of Hexavalent Chrome Cr\(VI\); \[Hexavalent Chromium Awareness CRSE- GE-EHS-425\] and will be assigned to you based on your job expectation.](#)
- 7.3 Perform visual inspection at the start of the activities and during their execution to verify the absence/presence of off-white to yellow contamination avoiding its contact with eyes and skin. If contamination is discovered, work should be halted on the task until the controls, can be put in place. It should be noted that the contamination in-situ, typically the baked-on residue, does not present, based on Industrial Hygiene monitoring conducted on various outages, present a significant airborne hazard until it is actively disturbed, by wiping etc.

8. GENERAL CONTROL MEASURES

- 8.1 If any suspicious residue is observed in a gas, steam & Aero units or associated pipework **STOP WORK** until you can initiate the following specific measures:
 - Contact your manager and/or EHS representative for additional guidance and assistance.
 - Avoid skin and eye contact with the residue. Always wear disposable nitrile gloves and eye protection whenever inspecting or handling parts potentially contaminated with any suspicious residue. If your clothing has the potential to become contaminated wear protective disposable coveralls (e.g., Tyvek or similar Type 5/6).
 - Parts contaminated with Cr(VI) should be cleaned prior to any work that might disturb the residue (e.g., manual or wire brushing, welding, brazing, grinding, sanding, cutting, scraping or scoring etc.) as this may cause the residue to become airborne. If this is not possible for technical reasons, then a face shield, goggles and proper respiratory protection (FFP3/P99 mask) should

be worn in addition to the PPE mentioned above. Liaise with your EHS team to ensure that the controls are sufficient and there is no potential for the contamination to become airborne and have potential to contaminated and adjacent work areas.

8.2 If work may cause the residue to become airborne the following measures must be implemented:

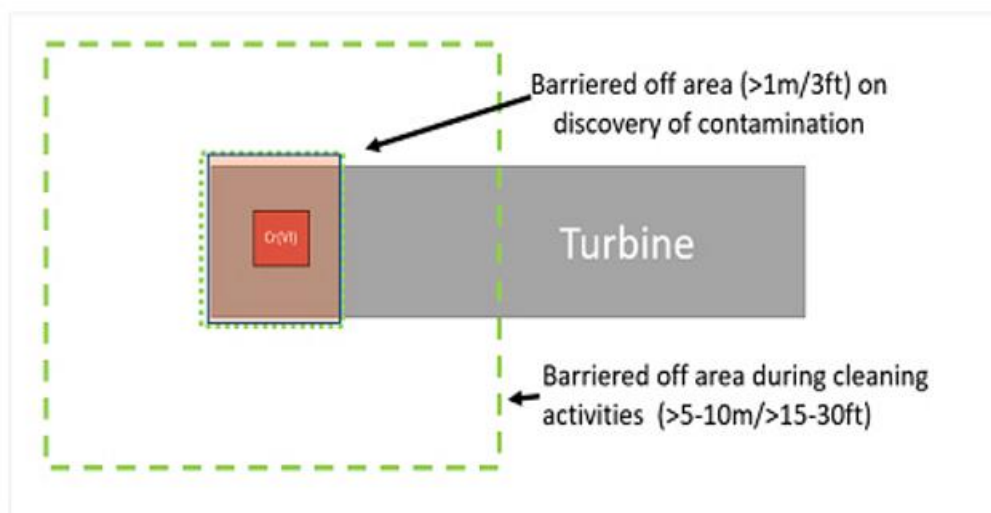
- The area shall be demarcated, and access controlled until completion (see Section 8 Exclusion Zone).
- NEVER use compressed air, which may cause the residue to become airborne.
- WD-40 or Kroll Oil should be applied to dampen affected areas prior to removal by scraping and/or wiping. EXCEPT when:
- Contamination has been found under Insulation (Glasswool etc) in this case water or water with mild detergent on damp rags/cloths, minimal water and allow it to dry will avoid potential for rusting. (WD40 etc should not ne used as it presents a fire risk once insulation is reinstated).
- While conducting cleaning work, additionally, wear a face shield or goggles and proper respiratory protection (typically FFP3/P99 mask) in addition to disposable coveralls and nitrile gloves.
- Always observe good hygiene practices; no eating, drinking, smoking or bathroom breaks should be taken without first washing hands and face, or any other potentially exposed area, to avoid the risk of accidental ingestion of material.

8.3 If potentially contaminated component parts are to be moved, take precautions to minimize direct contact or release of the residue during handling and transportation. All such parts should be placed in plastic bags or securely wrapped in plastic and marked to indicate that Cr(VI) hazard may be present. It is required to decontaminate parts prior to reinstatement of transport offsite, e.g., to a service center. (see section 12 for further information)

9. EXCLUSION ZONE

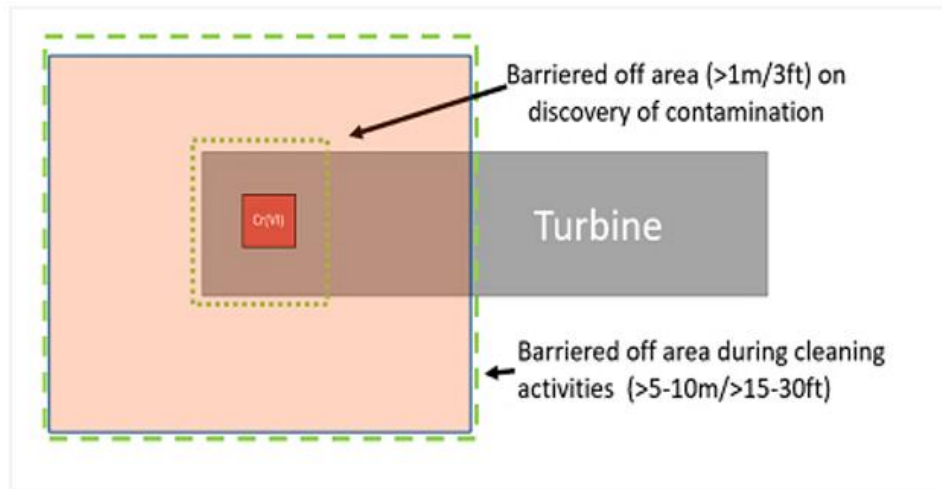
9.1 As part of the risk assessment, if contamination is identified and there's the potential for workers to come into contact with the contamination, and exclusion zone/area should be demarcated, to avoid accidental contact or exposure.

9.2 An initial barrier should be set-up in a way so that it protects any worker from disturbing the contamination accidentally with a body part or with any equipment, typically this should be greater than 1 m/3ft.



9.3 When cleaning / decontamination or disturbance activities where contamination has the potential to become airborne are being conducted a larger exclusion zone is likely to be needed. The barrier should

be extended typically > 5-10 m (15-30 ft). this is to protect other (unprotected) workers on the site not involved from inadvertent exposure to Cr(VI) while the residue is being disturbed. Other works can continue as normal outside the barriered area. Adjacent workers should be made aware of Cr(VI) being on the worksite and not to cross.



9.4 The following table resumes the minimum requirements for an exclusion zone.

Exclusion Zone			
Barriers	Entry Requirements	Other	PPE requirements
Caution – DO NOT ENTER barrier tape, cones, and signs to define the exclusion zone. Post signs in area – CAUTION: Cr(VI) Hexavalent Chromium Work Area	Entry is restricted to personnel with Hexavalent Chromium Awareness training and with knowledge of the safety plan and PPE/RPE qualifications.	Post Exclusion Zone Signs at entrance to the zone. No smoking, eating, or drinking is allowed in this area.	Disposable nitrile gloves and eye protection. Tyvek (or similar) disposable coverall if clothing has the potential to become contaminated. Unless the material is going to be physically disturbed, respiratory protection doesn't need to be worn otherwise FFP3/P99/P100 mask shall be used. A face shield should be used when wet/dampening methods are used to remove the contamination.

9.5 Additional **exclusion zones** should be identified in the areas in which contaminated parts are cleaned if not in-situ cleaning. These zones shall comply with the same minimum requirements reported above.

9.6 The following procedure describes how to EXIT the exclusion zones after finishing activities that can cause the contaminated residue to become airborne, typically this would be decontamination cleaning activities. [If entering the exclusion zone for a visual inspection, and not disturbance of the potentially contaminated material required is, no special EXIT procedure need to be followed].

- Remove helmet, wipe helmet with damp cloth.
- If shoes covers used, place them in a plastic bag for disposal. If not used, clean shoes with a damp cloth. Put the used cloth/rag in a plastic bag for disposal.
- Remove the Tyvek coverall by carefully rolling it inside out and place it in a plastic bag for disposal
- Remove outer gloves if worn (eg leather gloves) and place it in a plastic bag for disposal.
- If needed, keep the respirator on during the processes listed above.
- If needed remove the respirator. If disposable type place in waste. If reusable type with cloth, and dry, then place in designated containers.

- Remove inner gloves (nitrile gloves) taking care to ensure that any potential contamination does not make contact with hand skin.
- 9.7 After leaving the exclusion zone, proceed directly to the established wash area to thoroughly wash hands/wrist and face or any potential exposed skin with water and soap.
- 9.8 After completing all work activities, clean the tools and the work area with appropriate method, to ensure that there is not contamination remaining. Do not return tool to tool container with potential contamination.
- 9.9 All contaminated, consumable and used PPE/RPE should be disposed of in appropriated waste containers for marked as Hexavalent Chromium Cr(VI) Hazard.

10. PERSONAL PROTECTIVE EQUIPMENT

10.1 In addition to general PPE always required when working and handling materials (Helmet, Safety Shoes, Work Clothing, Gloves and Eye Protection) the following PPE shall be used when entering a limited containment restricted access zone or in case of a potential exposition to contaminated parts:

10.2 Disposable coveralls

- Disposable coveralls are needed if there is a potential for Cr(VI) contamination on clothing.
- Disposable Tyvek coveralls or similar Type 5/6 coverall.
- If work involves no physical disturbance of contamination, Tyvek coveralls and booties are not normally required.



10.3 Overshoes:

- The requirement for overshoes is only required for decontamination activities.
- Considering the use of overshoes may bring loss of sole grip and could represent a slipping hazard when used on ladders make sure hazards are considered on your job hazard analysis.



10.4 Gloves:

- A double layer of disposable nitrile gloves shall be used while handling contaminated parts
- Single only nitrile gloves if used under cut resistant gloves or other type of gloves



10.5 Respiratory Protection (RPE)

Work NOT disturbing residue

- Due to the nature of the contamination, i.e., "baked on" UNLESS the material is going to be physically disturbed, a respirator **does not need to be worn** as the material will not become airborne. (examples: working near the material, inspections, moving parts, etc.)



Work disturbing residue

- Respiratory protection is needed if the residue is be physically disturbed, a half face respirator with P3 or P99/100 particulate filter shall be used. Face mask can be single use disposable type FFP3 (e.g. 3M 8233) or silicon mask type & P3 or P99/100 cartridge. Workers should be clean shaven. In this case workers need information about the correct use of the mask assigned to them.



IF Using wet/dampening cleaning method methods for removal of residue; a face shield should be used in addition to safety glasses. If a face shield not available googles can used or a full face respirator.



11. CLEANING

	TITLE:		DOCUMENT NUMBER:			
	Hexavalent Chromium Cr(VI) Contamination EHS Plan		0807-G1			
	REGION:	Global	REVISION:	4.0	DATE:	MAR-2026
					PAGE:	9 of 12

11.1 **Use a wet method when cleaning all contamination** this is achieved by spraying/misting contaminated surfaces with a suitable oil-based wetting agent (e.g. Kroil Oil or WD-40) before scrapping or wiping off. Keep affected area damp to prevent material from becoming airborne, this may require additional spraying with oil as the cleaning progresses. **While cleaning / decontamination jobs are taking place, make sure a secondary containment, or material is used to prevent liquid contacting the floor.**

- NOTE: IF Contamination has been found under Insulation (Glasswool etc) in this case water or water with mild detergent on damp rags/cloths with minimal water used to avoid rusting. (WD40 etc should not be used as it presents a fire risk once insulation is reinstated).

11.2 A component part is considered clean/decontaminated when it has been wiped down with a damp rag and there is no visible residue remaining.

11.3 It should no longer present a hazard from Cr(VI) to any other subsequent work involving the part(s).

11.4 Part should be cleaned prior, to be shipped off site, as contaminated part poses a risk to other workers, at workshops etc (see Section 12 for instruction on this).

12. DECONTAMINATION BEFORE HOT WORK

12.1 The form of the Cr(VI) contamination being found is a baked-on surface residue.

12.2 The controls needed for Hot Work and Cr(VI) welding, torching, etc. are more complex (air movers, local exhaust, etc.), due to the difficulty of controlling airborne fume.

12.3 Before any hot work is performed in an area with Cr(VI) contamination, the material should be cleaned off to avoid the generation of additional Cr(VI) fumes being welding fumes.

13. CLEANING/DECONTAMINATION OF TURBINE PARTS DURING OUTAGES PRIOR TO SHIPPING OFFSITE TO GE SERVICE CENTRES OR 3RD PARTIES

13.1 The requirement is now that visible contaminated parts be cleaned & decontaminated onsite prior to shipping. This is irrespective if this is being initiated by FieldCore or the customer themselves.

13.2 Shipping contaminated parts, potentially presents an unacceptable risk to Service Centre and/or 3rd Party Vendors, workforce.

Some of whom may not have the knowledge and/or training to deal with the Cr(VI) hazard, adequately.

13.3 Actions Required:

Use the wet method to clean (e.g. WD40/Kroll Oil), dispose of any contaminated PPE correctly and bag/label the parts accordingly.

13.4 After cleaning with the wet method (e.g. WD40/Kroll Oil) and if no yellow is visible, you can ship parts as normal shipments. There is no need for any further testing, a visual inspection is sufficient.

IF part(s) DO NOT require further disassembly off site no further action

If the parts, require further disassembly, eg at a Service Centre, the shipped parts, should be cleared marked,

Decontaminated of Hexavalent Chromium Cr(VI). Further disassembly, has possibility of further contamination in previously inaccessible areas, PLEASE USE CAUTION!

This will enable the receiving center to prepare for potential Cr(VI) contamination within internal of parts

14. APPLICATION OF ANTI SEIZE

14.1 As a result of testing performed as part of a RCA (Root Cause Analysis), it was found that some anti-seize compounds that contain calcium have the potential to result in the formation of Cr(VI) when applied to chromium containing materials and exposed to high temperatures.

14.2 The following anti-seize (via GE Gas Power Product Safety are obsolete and should **NOT** be ordered or used;

Part No.	Name	Manufacturer	Manufacturer Part No	Status
P001	Heavy Duty Anti-Seize (1.2lb can) (formerly C102)	Loctite	51607	OBSOLETE Do NOT Use
P002	N-7000	Loctite	51273	OBSOLETE Do NOT Use
P003	Molykote P-37	Dow Corning	-	OBSOLETE Do NOT Use
P004	Heavy Duty Anti-Seize (8lb can) (formerly C102)	Loctite	36354	OBSOLETE Do NOT Use
P005	T8E-G	Tiodize AC&C	-	OBSOLETE Do NOT Use

14.3 This assessment also identified two approved anti-seize compounds which did not indicate any appreciable production (<1 ppm) of Cr(VI) in this test environment.

14.4 It is recommended to use the following commercially available anti-seize compounds that did not show any appreciable production of Cr(VI) in a test environment and do not contain any CMR (Carcinogenic, Mutagenic, Reprotoxic) registered substances.

These anti-seize products by Service line for use on outages:

Gas For use on Frame 5, 5-2, 6E, 6F, 7E, 7F, 7H and 9E, 9F, 9H

- Klüber Paste HEL 46-450 (HTGD481614P...)
- Bonderite L-GP GP460

For use on GT8, GT9, GT11, GT13, GT24, and GT26 gas turbines.

- Klüber Paste HEL 46-450 (HTGD481614P...)

Steam

- Klüber Paste HEL 46-450 (HTGD481614P...)
- Bonderite L-GP GP460
- Loctite LB N-1000

Aero

- Klüber Paste HEL 46-450 (HTGD481614P...)
- Bonderite L-GP GP460 (Aero)

Others with approval from GE Vernova: Refer to Section 2 of [GEV Engineering: 287A1397](#)

A 50:50 Graphite-Petrolatum compound conforming to SAE AMS2518 or MIL-T-5544 and may be substituted by any other manufacturer's product which also meets the requirements of those specifications with review and approval of Design Engineering and Materials & Process Engineering. The following are known acceptable alternates which have been approved:

Acheson GP 460 • Loctite Graphite50 (LB 51084) • Royco 44 • NYCO Grease GN GA 47
 • Ruscoe Lub Tork, • Armite SAE-AMS 2518

Implement a process that does not allow unapproved anti-seize compounds to be ordered and brought on- site.

- [EHS Alert Use of prohibited anti-seize on Outages \[Cr\(VI\) Formation\]](#)
- [PSSB 20180709A-R4](#)

15. PERSONAL HYGENE

- 15.1 Always observe good hygiene practices. Cleaning skin with soap and water, after working with Cr(VI) contamination. If skin is accidentally exposed to contamination, eg a nitrile glove splits, or worker touched their face, it should be cleaned promptly.
- 15.2 NO eating, NO drinking, NO smoking, phone calls or bathroom breaks should be taken without first washing hands and face or any other potentially exposed skin area to avoid the risk of ingestion.
- 15.3 Washing facilities, (normal soap and water is sufficient) shall be provided for removing Cr(VI) from the skin and affected employees shall use these facilities when necessary.

16. HOUSEKEEPING

- 16.1 Housekeeping is particularly important in work areas where activities on Cr(VI) contaminated parts are being conducted that can cause the contaminated residue to become airborne. Take all practicable measures to prevent the material to become airborne. Clean up accumulated residue by wet methods WD40/(Water for where insulation will be reinstated) & damp rag.

17. PROHIBITED ACTIVITIES

- 17.1 The following activities are prohibited, as they present an unacceptable risk. Other prohibited activities may be specified. Personnel engaging in prohibited activities may be subject to disciplinary action in accordance with established FieldCore policies.
- 17.2 Do not eat, drink, smoke, chew gum or tobacco, or apply cosmetics, barrier creams, suntan lotion etc within the Cr(VI) controlled. As these activities can provide a path for Cr(VI) to enter the body.
- 17.3 Do not use compressed air, abrasive or dry ice (CO₂) blasting on contaminated parts.
- 17.4 Unless no other option is available, any other method of aggressive cleaning (e.g., wire brushing, using rotary power tools, etc.) should be discouraged. Use wet methods (WD40/rags etc and water for contamination under insulation) for cleaning whenever possible. All efforts should be made to remove as much contamination as possible first before any work begins (e.g., from bolt threads).

18. ENVIRONMENTAL ASPECTS - WATER

- 18.1 All part should be decontaminated with the wet oil method, BEFORE any secondary methods are use, the parts should be visibly clean.

Washing contaminated parts with water (e.g. Power washer etc) can cause an uncontrolled discharge of Cr(VI) into the environmental (ground, drains, water bodies, etc.) so the following precaution shall be implemented:

- 18.2 If a washing machine is used to clean the parts, it shall have a closed water circuit.
- 18.3 Water discharges shall be collected and disposed as liquid waste according to local legislation.
- 18.4 Water treatment must be guaranteed before any potentially contaminated water discharge. i.e., waste water or similar should not be release into drains that do go for treatment.
- 18.5 Also leaching by rainwater could cause a contaminated water discharge. To avoid this the following measures shall be implemented:
 - Protection from rain and weather all contaminated parts if they are placed outside by using plastic sheeting, tarpaulin, or similar.

19. ENVIRONMENTAL ASPECTS - AIR

- 19.1 Precaution must be implemented to avoid the air dispersion of the contaminated dust. General control measures below have been shown to effectively do this :
 - WD-40 or Kroll Oil (Mild detergent & Water for contamination under insulation should be applied to dampen affected areas prior to removal by scraping and/ rubbing or wiping off.
 - NEVER use dry methods for removal especially compressed air or blasting techniques abrasive/CO₂, which will cause the residue to become airborne, and is an unacceptable risk to worker health and the environment .
 - Protect contaminated parts from wind and cover them with plastic sheets if necessary.

	TITLE:		DOCUMENT NUMBER:			
	Hexavalent Chromium Cr(VI) Contamination EHS Plan		0807-G1			
	REGION:	REVISION:	DATE:	PAGE:		
	Global	4.0	MAR-2026	12 of 12		

20. ENVIRONMENTAL ASPECTS - WASTE

20.1 All the following shall be considered as Special Hazardous Wastes:

- Turbine parts piping and debris with a potential contamination by Cr(VI);
- PPE used in the restricted access zones;
- Plastic and polyethylene sheeting, cover and bags used to protect contaminated parts; to cover floors and to protect from contamination;
- Cleaning Supplies and debris – material collected by HEPA vacuums, cloths, and rags used for wiping surfaces.

20.2 All wastes potentially contaminated with Cr(VI) shall be:

- Considered as potentially hazardous waste
- Collected in adequate container cleared marked to indicated Hexavalent Chromium Cr(VI) Waste (e.g. in 0.6 mil polyethylene bags double-bagged. Place the bags in, metal or plastic drums, or waster container, which are closed and secured at the end of work each day):
- Classified according to local Waste Codes or similar. The customer should aware that Special Waste has been generated, so they can arrange appropriate disposal.

20.3 All waste must be disposed according to local legislation relating to Cr(VI) waste.

21. ATTACHMENTS

21.1 GEV PSSB & Fieldcore CoreFacts Relating to Chromium Contamination

- [PSSB 20180709A-R4 Observation of Cr\(VI\)](#)
- [R6 - Contaminated Components - Service Centers \(09.2021\)](#)
- [Cr\(VI\) Update \(03.2020\)](#)
- [CoreFact R2 \(07.2018\)](#)

21.2 Cr(VI) Toolbox for use on site as training and awareness [Cr\(VI\) Toolbox v3 04.26](#)

21.3 Risk Assessment Flowchart v2: [Cr\(VI\) Outage - Risk Assessment - Flowchart v2](#)

21.4 Site Sampling: [Field Test for Hexavalent Chromium Cr\(VI\) FieldCore](#)

21.5 Cr(VI) Contamination “Response” Kit [Suggested content of Cr\(VI\) Kit for Tool containers](#)